

Table 2: Python-based libraries for data engineering and visualisation

Library	Output	Category	Key applications
Lux	Inline plots	Auto visualisation	Smart insights
Matplotlib	Static images	Base plotting	Core library
Seaborn	Static images	Statistical plots	Pretty charts
Plotly	Web plots	Interactive visualisation	Rich widgets
Bokeh	Browser output	Interactive plots	Dash ready
Altair	Web charts	Declarative visualisation	Grammar spec
Holoviews	Static/Interactive	High-level visualisation	Auto plots
hvPlot	Static/Interactive	Easy plots	Quick API
GeoPandas	Maps	Geo plots	GIS tools
Folium	Leaflet maps	Map visuals	Web maps
Cartopy	Static maps	Geospatial plots	GIS support
PyViz Panel	Web apps	Dashboarding	Multi-library
Dash	Web apps	Interactive apps	Plotly based
Pandas Plot	Static charts	Quick visualisation	Built-in API
Yellowbrick	Model plots	ML visualisation	ML diagnostics
Pygal	Vector output	SVG charts	Clean SVG
Pycharts	Web charts	ECharts visualisation	China charts
VisPy	High-performance	GPU visualisation	OpenGL-based
Mayavi	Scientific 3D	3D visualisation	VTK tools
Plotnine	Static images	Grammar charts	ggplot style
Datashader	Scalable visualisation	Large data	Huge datasets
K3D	Jupyter 3D	3D plots	WebGL based
ipyvolume	Jupyter 3D	3D widgets	Interactive 3D
PyVista	Static/Interactive	3D models	Mesh tools
VTK	Scientific visualisation	3D engine	Heavy duty
Manim	Vector video	Math animation	Animation library
Matplotlib	Static images	Pretty themes	Style boost

Lux integrates various features in Python including automatic visualisation suggestions, intelligent data profiling, minimal code usage, Pandas DataFrame integration, toggle visualisation mode, interactive visual widgets, chart recommendation

engine, quick summary stats, outlier identification support, Jupyter notebook compatibility, built-in analytics insights, correlation detection tools, user intent specification, export visualisation options, attribute-based exploration,

faceted visual comparisons, visual filtering interface, multi-chart generation, and many others. All these features enable highly effective visualisation in data engineering and analytics-based projects.

Lux integrates the following functions for plotting and

advanced visualisation:

- Intent specification for statistical data analytics
 - Metadata analysis
 - Faceted chart generation
 - Correlation analysis and views
 - Relationship views
 - Recommendation sections with statistics
 - Querying and filtering
- You can use the following code to install Lux in Google Colab:

```
!pip install lux-api
from google.colab import output
output.enable_custom_widget_manager()
```

LUX Python Code for Visualization of Sample Dataset in Script

```
import lux
import pandas as pd
# Sample Dataset for Visualization
```

Toggle Pandas/Lux					
	Product	Month	Sales	Profit	Units_Sold
0	Laptop	January	220	30	10
1	Laptop	July	250	45	19
2	Smart Watch	January	220	50	10
3	Smart Watch	July	250	90	23
4	Tablet	July	90	15	6
5	Tablet	January	90	20	9
6	IoT Gadget	January	165	20	13
7	IoT Gadget	July	180	45	15

Figure 3: DataFrame with option to toggle with Lux